MosyMamba: Modal-Synergy Mamba Network for Hyperspectral and Multispectral Image Fusion

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***Abstract*—Hyperspectral and multispectral image fusion aims to overcome the hardware limitations of the imaging system by fusing low-resolution hyperspectral images (LR-HSIs) with high-resolution multispectral images (HR-MSIs) to generate high-resolution HSIs (HR-HSIs). Existing vision Mamba models mainly perform spatial sequence modeling, which insufficiently exploits the rich spectral information of LR-HSI and limits its effective interaction with the spatial details of HR-MSI. To address this, we propose a novel modal-synergy Mamba network for hyperspectral and multispectral image fusion (MosyMamba) in this article. In particular, we design an efficient cross-modality interaction Mamba block (CIMB), wherein a synergistic scanning mechanism facilitates detailed interaction between the spectral information of LR-HSI and the spatial geometry of HR-MSI. This approach enables comprehensive contextual extraction and pixel-level spatial–spectral understanding. Furthermore, we introduce a difference-perceptive fusion module (DPFM) with an adaptive feature difference-aware fusion (FDF) strategy to maintain spatial–spectral consistency. The module perceives the unique information required by spatial and spectral features and dynamically supplies crucial details for them, thereby enhancing the model performance effectively. Extensive experiments demonstrate the superiority of the proposed method over state-of-the-art (SOTA) approaches across three widely used datasets, e.g., achieving a 1.9027 dB gain in peak signal-to-noise ratio (PSNR) and a 20.58% improvement in spectral angle mapper (SAM) on the CAVE dataset.**

***Index Terms*—Difference-perceptive, hyperspectral image (HSI), image restoration, Mamba, multispectral and HSI fusion.**

I. Introduction

HYPERSPECTRAL images (HSIs) usually have dozens to hundreds of spectral bands. Thus, they can highlight subtle spectral differences among objects and capture more detailed scene information compared to RGB images. This advantage allows HSIs to support various image analysis applications, such as scene classification [1], [2], [3], [4], [5], object detection [6], [7], [8], [9], [10], ecological monitoring [11], [12], [13], [14], [15], etc. However, the physical limitations in imaging technology lead to HSI degradation in practical applications. To ensure spectral resolution, hyperspectral sensors often reduce spatial resolution, causing a loss of spatial details. This tradeoff impacts visual quality and limits analysis. Hyperspectral and multispectral image fusion (HS/MS fusion) has become a widely used method to obtain high spatial resolution HSIs. It integrates high-resolution multispectral images (HR-MSIs) with low-resolution HSIs (LR-HSIs) to generate high-quality HSIs.

Deep learning-based approaches have been increasingly applied to HS/MS fusion, demonstrating impressive recovery capabilities. Initially, convolutional neural networks (CNNs) are employed for HS/MS fusion methods [16], [17], [18], [19], [20], learning the end-to-end mapping between LR-HSI and HR-MSI. Recent studies further explore CNN-based or CNN-assisted architectures for blind and unsupervised HS/MS fusion, such as the hybrid network of Transformer and CNN [21]. However, CNNs focus on local features captured by convolutional kernels, lacking in building long-range dependence and thus limiting the model's representational capacity. Subsequently, Transformer architectures achieve significant success in HS/MS fusion due to their outstanding performance in capturing long-range dependencies, overcoming the limitations of CNNs. Fusformer [22] first proposes the application of Transformers to the HS/MS fusion task, PSRT [23] proposes a novel pyramid shuffle-and-reshuffle Transformer, and BDT [24] applies dilation transformers from spatial and spectral perspectives, respectively. MSST [25] extracts spatial and spectral information separately at multiple scales. Recently, MIMO-SST designs a multi-input multi-output spatial–spectral Transformer to progressively recover spatial structures and spectral details for HS/MS fusion [26]. However, existing methods still face limitations in fully exploiting the complementary information between LR-HSI and HR-MSI. Moreover, these methods fail to consider the differences between features extracted by spatial and spectral extractors, which proves significantly important in feature fusion in our study.

In summary, although existing methods achieve some success, they still face the following issues: *1) Deficient Feature Fusion:* they individually extract features from the spatial domain or spectral domain and leave the requirement of feature integration to the following modules, leading to deficient integration of LR-HSI and HR-MSI information and *2) HR-MSI Feature Dominance:* we found that the HR-MSI branch may dominate the subsequent feature fusion process in local high-frequency regions in traditional fusion modules, while the LR-HSI branch contributes less. This suggests that the network fails to effectively utilize hyperspectral features, leading to poor reconstruction quality in the channel dimension.

Received 16 March 2026; revised 11 May 2026 and 16 June 2026; accepted 10 July 2026. Date of publication 20 July 2026; date of current version 31 July 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 92470108 and Grant 62272363 and in part by the Joint Laboratory for Innovation in Satellite-Borne Computers and Electronics Technology Open Fund under Grant 2024KFKT001-1. *(Corresponding author: Jie Guo.)*

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Digital Object Identifier 10.1109/TGRS.2026.3715225

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Inspired by the latest advancements in the Mamba model and its strong global context modeling [27], [28], we propose a novel dual-branch modal-synergy Mamba network for hyperspectral and multispectral image fusion (MosyMamba) by learning complementary spatial and spectral information. In particular, we design a cross-modality interaction Mamba block (CIMB), which excels at integrating the corresponding spatial regions and spectral bands of LR-HSI and HR-MSI. Considering that traditional vision Mamba ignores the spectral dimension information, this block enhances local spectral detail representation by integrating the spectral information of surrounding pixels into the original state equation. Moreover, an efficient sampling and sequence grouping scanning mechanism is proposed. This method reduces the redundancy and delays caused by scanning the entire feature map in traditional vision Mamba. It uniquely captures the connectivity and spectral correlation between local and nonlocal pixels in both modalities. In this way, each state update simultaneously receives information from both modalities and is explicitly guided by local spectral discrepancy. Furthermore, as analyzed in Fig. 5, HR-MSI features may dominate reconstruction in conventional fusion modules, weakening the contribution of LR-HSI's spectral cues. To mitigate this issue, we propose a difference perceptive fusion module (DPFM) with an adaptive feature difference-aware fusion (FDF) strategy to better integrate the information from both modalities. The module utilizes neighborhood difference descriptions and similar-region assistance, focusing not only on local differences but also on distant potential similarities, therefore enhancing the network's understanding of detailed differences in regions. During training, we adopt spatialwise and channelwise L_1 losses, respectively, for two modalities. These loss objectives continuously sharpen details, ensuring that the fused high-resolution HSI (HR-HSI) image maintains spatial–spectral consistency. The proposed MosyMamba achieves superior performance over the state-of-the-art (SOTA) methods shown in Fig. 1 while maintaining a compact model.

Our contributions are summarized as follows.

1) We propose MosyMamba, a modal-synergy selective-state space model (SSM) network for HS/MS fusion that generates high-fidelity HR-HSI by jointly exploiting LR-HSI's spectral richness and HR-MSI's spatial geometry.
2) We design a novel Mamba scanning mechanism that efficiently captures spatial and spectral characteristics from both short- and long-range modeling, ensuring robust interaction between LR-HSI's spectral and HR-MSI's spatial distributions. Besides, our proposed FDF strategy effectively enhances the network's ability to perceive the information required by each modality, thereby boosting performance.
3) Extensive experiments demonstrate that our MosyMamba significantly outperforms SOTA methods on HS/MS fusion tasks across diverse HSI datasets.

The remainder of this article is organized as follows. In Section II, we briefly review recent methods related to our approach. In Section III, we introduce the detailed architecture of the proposed MosyMamba. Section IV presents experimental results and analysis on the CAVE, Harvard, and Chikusei datasets. Finally, we conclude this article in Section V.

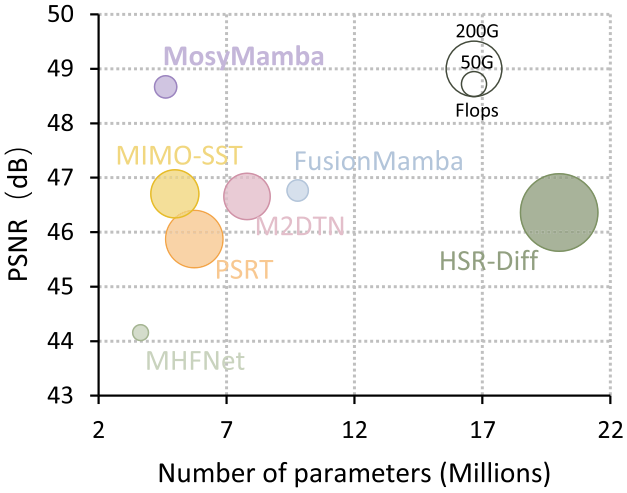


Fig. 1. PSNR-parameters–flops comparisons with other HS/MS fusion methods. Our MosyMamba surpasses other methods while requiring significantly cheaper flops and parameters.

II. Related Work

A. LR-HSI/HR-MSI Fusion Methods

Traditional HS/MS fusion methods [29], [30], [31], [32] fall into two categories: pansharpening algorithms and decomposition-based methods. Pansharpening methods include multiresolution analysis (MRA), component substitution (CS), and variational optimization. MRA methods [33] inject high-resolution spatial information from HR-MSI into LR-HSI via multiresolution decomposition. CS methods [34] transform LR-HSI to extract spatial and spectral components, replacing the spatial component with HR-MSI. Variational optimization methods [35] tackle the inverse problem of image degradation using variational theory and iterative algorithms. However, they heavily depend on MSI-HSI correlation, risking information loss from spatial–spectral discrepancies. Decomposition-based methods, such as coupled nonnegative matrix factorization (CNMF) [36] and coupled sparse tensor factorization (CSTF) [20], model the image cube as matrices or tensors. While these methods reduce information loss, they depend on precise sensor coefficients and manual priors, restricting their generalizability.

Benefiting from the robust capabilities of deep learning in computer vision, numerous deep learning-based methods have emerged for HS/MS fusion, showing promising results in fusing HR-HSI. CNN-based methods directly learn the highly nonlinear mapping between observed images and the target HR-HSI, integrating LR-HSI and HR-MSI to produce optimal HR-HSI. For instance, MHF-Net utilizes an iterative algorithm with embedded linear mappings to incrementally fuse HR-HSI from HR-MSI and LR-HSI [19]. HSRnet scales LR-HSI to the HR-MSI level to preserve spectral fidelity, integrating an attention module to improve spatial detail [37]. Although CNNs excel in capturing local features and translational invariance,

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they struggle with long-range dependencies and self-similarity priors. To address these issues, Transformer-based HS/MS fusion methods have been developed [38]. Fusformer, the first to introduce the Transformer architecture to HS/MS fusion, leverages self-attention to explore global feature relationships [22]. PSRT introduces a window-attention-based pyramid structure to facilitate detail extraction [23], and MSST-Net [25] incorporates a self-supervised pretraining strategy to boost performance and generalization. Recently, DCTransformer introduces a reciprocal fusion strategy to enhance the interaction and coordination between LR-HSI and HR-MSI [39]. MIMO-SST further designs a multi-input multioutput spatial–spectral Transformer to progressively recover spatial structures and spectral details for HS/MS fusion [26]. However, these methods fail to fully exploit the complementary information between LR-HSI and HR-MSI and suffer from computational complexity that scales quadratically with the number of patches.

III. Methodology

A. Preliminary

The SSMs initially serve natural language processing tasks [40], [41], [42]. The representative S4 model [41] effectively addresses long-range dependencies in sequence modeling while minimizing storage requirements. Building on this, Mamba [43] further optimizes context compression capabilities, enabling selective sequence information processing while maintaining training and inference efficiency. The model relies on a classical continuous system that maps a 1-D input function or sequence $x(t) \in \mathbb{R}$ to an output $y(t) \in \mathbb{R}$ by evolving an intermediate hidden state $h(t) \in \mathbb{R}^N$. The process can be expressed as a linear ordinary differential equation

$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t) \tag{1}$$

$$y(t) = \mathbf{C}h(t) + Dx(t) \tag{2}$$

$$\bar{\mathbf{A}} = \exp(\Delta\mathbf{A}), \bar{\mathbf{B}} = (\Delta\mathbf{A})^{-1}\left(\exp(\Delta\mathbf{A}) - \mathbf{I}\right) \cdot \Delta\mathbf{B} \tag{3}$$

where N denotes the state size, $\mathrm{A} \in \mathbb{R}^{N\times N}$ represents the state transition matrix, $\mathbf{B} \in \mathbb{R}^N$ and $\mathbf{C} \in \mathbb{R}^N$ are the projection matrices, and $D \in \mathbb{R}$ refers to the skip connection, with D set to 1 in this study. The parameters $\bar{\mathbf{A}}$ and $\bar{\mathbf{B}}$, derived using a zero-order hold discretization method, are particularly well-suited for HSI data, which inherently exhibit discrete characteristics.

Recently, researchers have explored the application of Mamba models in computer vision tasks. For instance, VMamba [28] adapts the selective SSM to 2-D image data by employing 2-D selective scanning along four routes, accommodating the nonsequential nature of images. Vision Mamba [27] introduces a universal visual backbone that compresses visual representations using bidirectional Mamba modules. While the outputs of these methods encompass global sequence information, extracting results from Mamba's final state poses challenges for HSI tasks with spatial sensitivity and spectral characteristics.

B. Overall Framework

The overview of MosyMamba is illustrated in Fig. 2(a). It incorporates an innovative dual-branch framework aiming at integrating the spatial and spectral information from the collected LR-HSI $\mathbf{X} \in \mathbb{R}^{C\times h\times w}$ and HR-MSI $\mathbf{Y} \in \mathbb{R}^{c\times H\times W}$ to generate a target HR-HSI $\mathbf{Z} \in \mathbb{R}^{C\times H\times W}$. Here, C, H, and W denote the number of spectral bands, heights, and widths of the HR-HSI. Given that HR-MSI is richer in spatial information and LR-HSI is richer in spectral information, it follows that $c \ll C$, $h \ll H$, and $w \ll W$. The process begins by feeding the upsampled LR-HSI and HR-MSI into the **CIMB**, which employs a synergistic scanning mechanism within the joint state space (JSS) block to extract spectral features as well as spatial local and nonlocal features. Next, these features are fused through the **DPFM**. It first obtains feature differences between two modalities and then refines them using neighborhood information and similar regions' information. Finally, as the network iteratively refines, both branches produce HSI outputs with high spatial–spectral fidelity. During inference, we average the outputs of the two branches as the final output of MosyMamba.

C. Cross-Modality Interaction Mamba Block

To fully preserve the rich spectral information of LR-HSI and the high-resolution spatial distribution of HR-MSI, inspired by cross-attention, we design a CIMB based on a spectral-assisted state equation and a unique synergistic scanning mechanism for the HS/MS fusion task. The structure of the CIMB is depicted in Fig. 2.

1) Spectral-Assisted State Equation: In the HS/MS fusion task, low-resolution pixels in LR-HSI may contain information from multiple land cover types. To effectively enhance spectral fidelity for target objects, it is essential to aggregate spectral information of similar objects or materials while maintaining spatial distribution. Mamba excels at modeling long-range dependencies, widely used for extracting global context and learning discriminative features in vision tasks. However, Mamba's state equation primarily considers spatial dimensions, often neglecting spectral information. To improve Mamba's local spectral modeling capability, we design a new Mamba formulation for Joint 2-D-Selective-Scan (JSS2D).

In the state equation of JSS2D, we integrate the spectral information of surrounding points with the current scanning point to more effectively capture the spectral relationships in the local region. The process is described by

$$\begin{aligned} h_k &= \bar{\mathbf{A}}h_{k-1} + \bar{\mathbf{B}}x_k \\ &\quad + \gamma\frac{1}{U}\sum_{u=1}^{U}\sum_{t=1}^{D}\left(\theta_{spe}^{t}\log\frac{\theta_{spe}^{t}}{\phi_{spe}^{u,t}} + \phi_{spe}^{u,t}\log\frac{\phi_{spe}^{u,t}}{\theta_{spe}^{t}}\right)x_k \\ y_k &= \mathbf{C}h_k + x_k \\ \text{with } \bar{\mathbf{A}} &= e^{\Delta\mathbf{A}}\ \bar{\mathbf{B}} = (\Delta\mathbf{A})^{-1}\left(e^{\Delta\mathbf{A}} - \mathbf{I}\right)\cdot\Delta\mathbf{B} \end{aligned} \tag{4}$$

where $x_k \in \mathbb{R}^{B\times M\times D}$ is the input sequence, B is the batch size, M is the length of the sequence, and D is the feature dimension. By evolving the parameter $\mathbf{A} \in \mathbb{R}^{N\times N}$, projecting parameters $\mathbf{B} \in \mathbb{R}^{B\times M\times N}$ and $\mathbf{C} \in \mathbb{R}^{B\times M\times N}$, the learnable intermediate hidden state $h \in \mathbb{R}^{B\times M\times N}$ maps the x'_k to the

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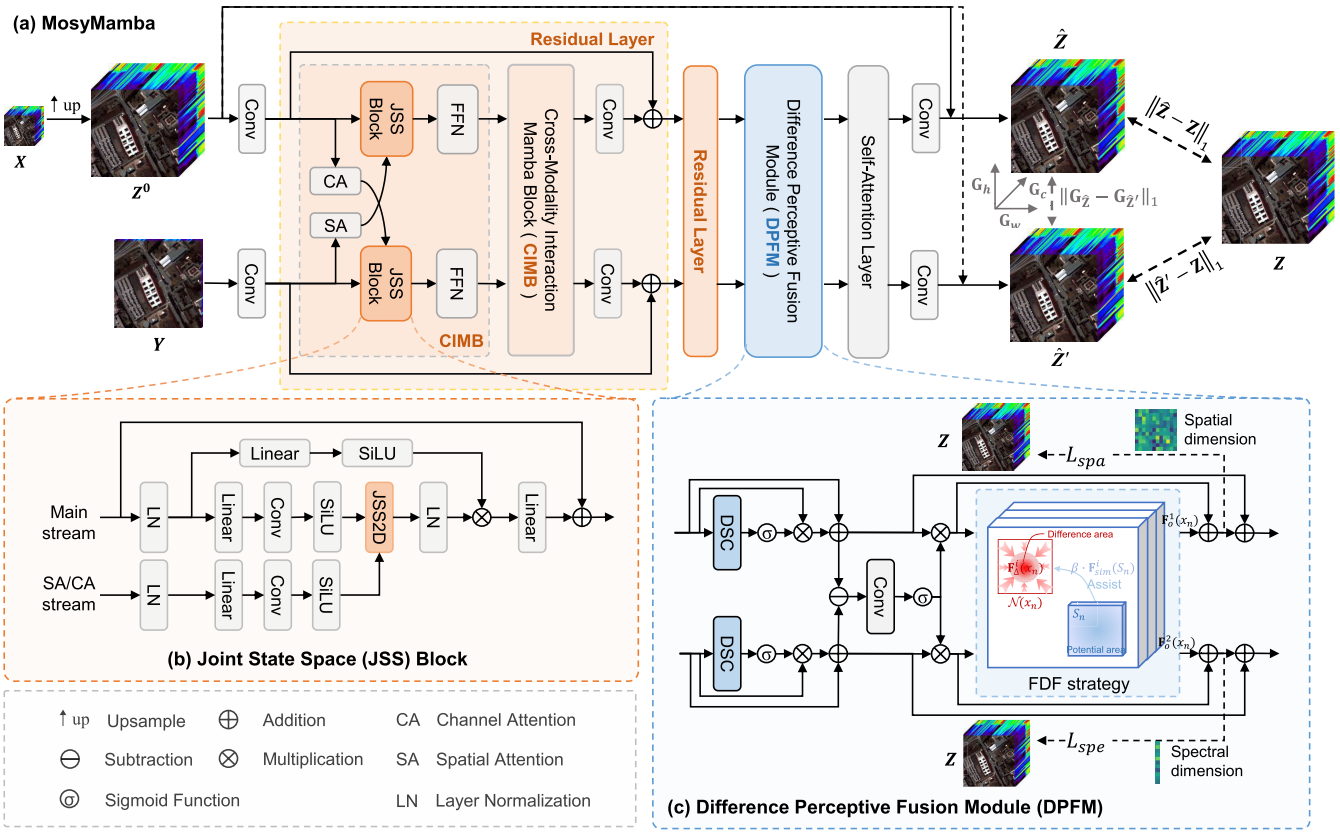


Fig. 2. (a) Overview of the MosyMamba for hyperspectral and multispectral image fusion. It consists of two residual layers containing CIMB, DPFM, and self-attention layers, focusing on their respective tasks. (b) JSS block. It differs from traditional vision Mamba by adding input from the SA/CA stream. (c) Specific process of DPFM. It applies the FDF strategy in two branch feature maps separately.

$y_k \in \mathbb{R}^{B\times M\times D}$, where N is the state expansion factor. $(\bar{\mathbf{A}}, \bar{\mathbf{B}})$ is the discretized version of $(\mathbf{A}, \mathbf{B})$ obtained through a zero-order hold [44]. $\Delta \in \mathbb{R}^{B\times M\times N}$ represents the discrete time step. $\mathbf{B}$, $\mathbf{C}$, and Δ are derived from the input sequence by calculating through a linear layer, with $\bar{\mathbf{B}}$ approximated by a first-order Taylor series expansion as $\bar{\mathbf{B}} \approx \Delta\mathbf{B}$. θ^t_{spe} and $\phi^{h,t}_{spe}$ represent the spectral distributions of the current pixel and the neighboring pixels, respectively. The bracketed term in (4) is a symmetric KL divergence between θ^t_{spe} and $\phi^{h,t}_{spe}$, which is nonnegative and measures local spectral discrepancy. By injecting this discrepancy into the state update, JSS2D allocates more modeling capacity to mixed or spectrally ambiguous regions, improving spectral fidelity in HS/MS fusion. Adjusting U expands the range of spectral information acquisition, accommodating targets of varying sizes. Here, we set it to 4. γ is used to adjust the weight and is set to 0.1.

2) Synergistic Scanning Mechanism: Traditional vision Mamba often processes the entire feature map during scanning, resulting in redundant information and significant delays. This redundancy mainly comes from the continuous scanning of adjacent pixels, which usually contain highly similar local information in HSI/MSI feature maps. To reduce such redundancy, we adopt an odd/even sampling strategy to separate adjacent pixels into two complementary subsequences. This avoids excessively consecutive processing of similar neighboring pixels while still preserving local spatial dependence through the one-pixel offset between odd and even positions. Since the two subsequences jointly cover the whole feature map, no spatial information is discarded.

As shown in Fig. 3, given feature maps $\tilde{\mathbf{X}}^{in} \in \mathbb{R}^{C\times H\times W}$, $\tilde{\mathbf{Y}}^{in} \in \mathbb{R}^{C\times H\times W}$, after flattening, we sample the odd and even positions from each feature maps and rearrange them into sequences $\tilde{\mathbf{Q}}_x$ and $\tilde{\mathbf{Q}}_y$ of length L. These sequences are then fused to form $\tilde{\mathbf{Q}}$

$$\mathcal{I}_k(L) = \{i \mid i \equiv k \,(\mathrm{mod}\, 2), 0 \leqslant i < L\} \tag{5}$$

$$\tilde{\mathbf{Q}}_x = Sort\left(\tilde{\mathbf{X}}^{in}_{flat}[:, \mathcal{I}_0(L)], \tilde{\mathbf{X}}^{in}_{flat}[:, \mathcal{I}_1(L)]\right) \tag{6}$$

$$\tilde{\mathbf{Q}}_y = Sort\left(\tilde{\mathbf{Y}}^{in}_{flat}[:, \mathcal{I}_0(L)], \tilde{\mathbf{Y}}^{in}_{flat}[:, \mathcal{I}_1(L)]\right) \tag{7}$$

$$\tilde{\mathbf{Q}} = \mathcal{H}\left(\tilde{\mathbf{Q}}_{x,i}, \tilde{\mathbf{Q}}_{y,i}\right), i = 0, 1, \ldots, L-1 \tag{8}$$

where $\mathcal{I}_k(L)$ denotes the index set, where k is the parity indicator: $k = 0$ denotes odd indices and $k = 1$ denotes even indices. $\tilde{\mathbf{X}}^{in}_{flat} \in \mathbb{R}^{C\times L}$ and $\tilde{\mathbf{Y}}^{in}_{flat} \in \mathbb{R}^{C\times L}$ represent the feature sequences of the flattened feature maps $\tilde{\mathbf{X}}^{in}$ and $\tilde{\mathbf{Y}}^{in}$, respectively. $Sort(\cdot,\cdot)$ is the sorting operation. $\mathcal{H}(\cdot,\cdot)$ denotes the token fusion operation consisting of convolution layer, where $\tilde{\mathbf{Q}}_{x,i}$ and $\tilde{\mathbf{Q}}_{y,i}$ are tokens at corresponding sequence positions. Sequentially scanning odd and even positions establishes dependencies with a one-pixel spatial offset, reducing redundancy. This design is equivalent to a stride-2 sampling along the flattened scan order. As neighboring pixels in hyperspectral scenes are highly correlated, stride-2 reduces repeated updates while still preserving local one-pixel-offset interactions.

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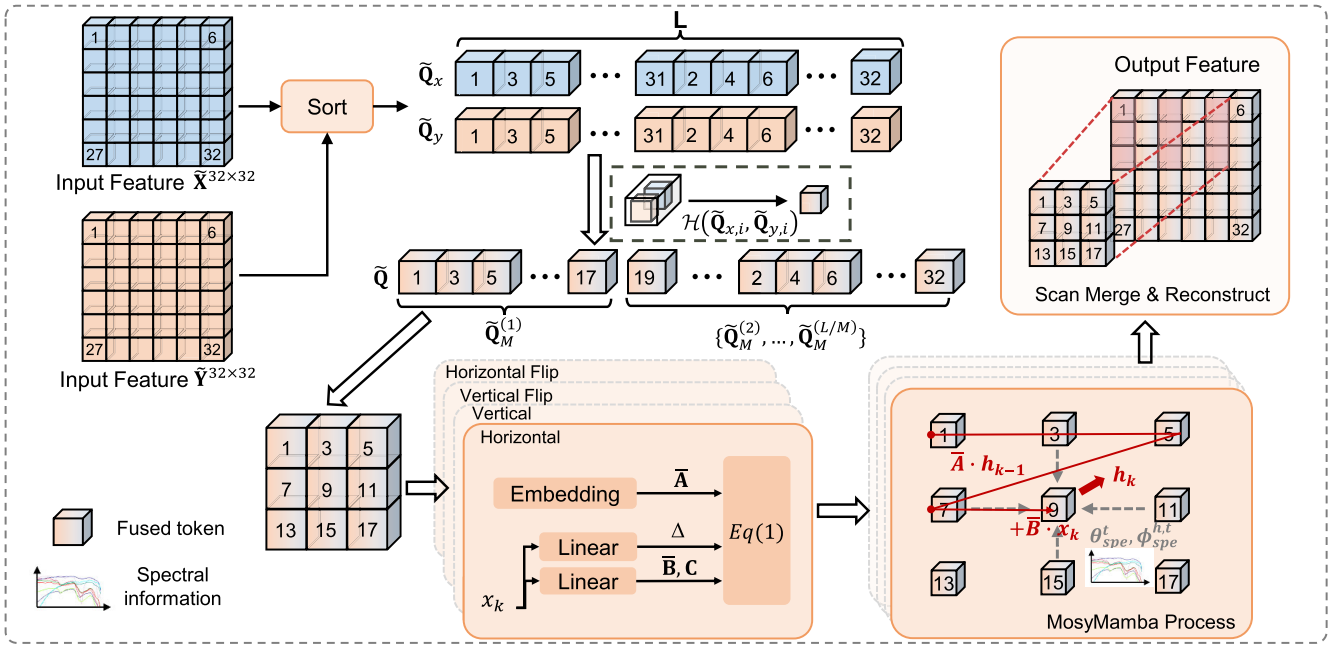


Fig. 3. Illustration of scanning methods in JSS2D using 32×32 feature maps. We incorporate spectral information into the state equation, enabling MosyMamba to capture the spectral details of neighboring pixels during the scanning process. We design a synergistic scanning mechanism, which enhances the interaction between LR-HSI and HR-MSI modalities while reducing redundancy and scanning delays.

Next, we divide the sequence of length L into d groups of sequences, each of length M ($d = L/M$). Notably, the first and second CIMBs in a residual Mamba layer are configured with 64 and 256 groups, respectively, to facilitate the capture of both relatively short-range and long-range information. This process is expressed

$$\tilde{\mathbf{Q}}_M^{(j)} = \left\{\tilde{q}_i \mid i \in \left[(j-1)M, jM - 1\right]\right\}, j = 1, 2, \ldots, \frac{L}{M}$$
$$\tilde{\mathbf{Q}} = \left[\tilde{\mathbf{Q}}_M^{(1)}, \tilde{\mathbf{Q}}_M^{(2)}, \ldots, \tilde{\mathbf{Q}}_M^{(\frac{L}{M})}\right]. \quad (9)$$

Then, each sequence group $\tilde{\mathbf{Q}}_M^{(j)}$ is expanded into four scanning paths and evaluated using the state equation.

D. Difference-Perceptive Fusion Module

The difficulty of feature fusion is to find an effective way of information fusion, and the difference feature is the most unique information source of the two output features of CIMB. We naturally focus our attention on the "conflict region" of the two features, tailoring an adaptive FDF strategy.

First, we utilize depthwise separable convolutions instead of traditional convolutions to reduce the computational cost. Then, independently extract the salient features of the input features

$$\tilde{\mathbf{Z}}^i_{\mathrm{DSC}} = \sigma\left(f_{\mathrm{DSC}}\left(\tilde{\mathbf{Z}}^i_{ap}\right)\right) \otimes \tilde{\mathbf{Z}}^i_{ap} + \tilde{\mathbf{Z}}^i_{ap}, \quad (i = 1, 2) \quad (10)$$

where σ is the sigmoid function and $f_{\mathrm{DSC}}(\cdot)$ is the depthwise separable convolution.

As shown in Fig. 2(c), for $\tilde{\mathbf{Z}}^i_{\mathrm{DSC}}$, we apply subtraction to extract the difference feature. After sigmoid activation, the corresponding difference feature weighting factor is

$$\rho = \sigma\left(\Delta_{diff}\mathbf{F}\right) = \sigma\left(\left|\tilde{\mathbf{Z}}^1_{\mathrm{DSC}} - \tilde{\mathbf{Z}}^2_{\mathrm{DSC}}\right|\right) \quad (11)$$

where $\Delta_{diff}\mathbf{F}$ denotes the difference feature. In the following, we multiply the weight ρ with $\tilde{\mathbf{Z}}^i_{\mathrm{DSC}}$, and improve the network's ability to pay attention to the difference feature regions by combining the neighborhood and potentially similar regions. Inspired by clustering concepts, we formalize this approach using the following operations.

1) Neighborhood Difference Description: Let the neighborhood $\mathcal{N}(x_n)$ of an arbitrary difference position x_n be initially set to a size of 4 × 4, and adaptively reduced during the network iteration to avoid the issue of prematurely focusing on the center, which could lead to a local optimum. The adaptive neighborhood difference feature at position x_n is given by

$$\mathbf{F}^i_{\Delta}(x_n) = \sum_{n\in\mathcal{N}(x_n)} f\left(\mathbf{f}_n, \mathbf{f}_c; \mathcal{N}(x_n)\right) \cdot (\mathbf{f}_n - \mathbf{f}_c)(\mathbf{f}_n - \mathbf{f}_c)^{\mathrm{T}} \quad (12)$$

where the function f is defined as

$$f\left(\mathbf{f}_n, \mathbf{f}_c; \mathcal{N}(x_n)\right) = \frac{1}{\varphi_{spa}\left(\|\mathbf{f}_n - \mathbf{f}_c\|_2^2\right)} + \omega_n \quad (13)$$

where $\mathbf{f}_n$ and $\mathbf{f}_c$ denote the features of neighboring points and different points, respectively. φ_{spa} denotes the mean operator over spatial axis. ω_n is an impact factor learned by training.

In HSIs, similar objects are often distributed across different regions of the image (e.g., the same type of vegetation or buildings). Although spatially distant, these objects share similar features. To address the limitations of purely local neighborhood descriptions, the FDF strategy leverages guidance from potentially similar regions. The following operation formalizes this approach.

2) Potentially Similar Regions Assist: Potentially similar regions are identified from the n nonoverlapping areas in the feature map, with their size determined by $\mathcal{N}(x_n)$ as defined in operation Section III-D1. Depending on the presence of a

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qualifying similar region $\mathcal{S}_n$, the difference feature is updated as follows:

$$\mathbf{F}_o^i(x_n) = \begin{cases} \mathbf{F}_\Delta^i(x_n) + \beta \cdot \mathbf{F}_{sim}^i(\mathcal{S}_n), & \mathcal{S}_n \ exists \\ \mathbf{F}_\Delta^i(x_n), & otherwise \end{cases} \tag{14}$$

where the $\mathcal{S}_n$ satisfies both the following spatial and spectral similarity conditions:

$$\begin{cases} \sum \|\mathbf{f}_n - \mathbf{f}_k\|_2 \leqslant \dfrac{1}{1+\sigma_{\Delta f}(x_n)} \\ \arccos\left(\dfrac{\mathbf{f}_n \mathbf{f}_k}{\|\mathbf{f}_n\|\|\mathbf{f}_k\|}\right) \leqslant \Omega \end{cases}, n \in \mathcal{N}(x_n) \text{ and } k \in \mathcal{S}_n \tag{15}$$

where $\sigma_{\Delta f}(x_n)$ denotes the standard deviation of feature differences within $\mathcal{N}(x_n)$. $\mathbf{F}_{sim}^i$ represents the features of the similar region. β and Ω are set to 0.1 empirically.

The feature map with different description obtained according to operation Sections III-D1 and III-D2 can be expressed as

$$\tilde{\mathbf{Z}}_{\text{DIV}}^i = \tilde{\mathbf{Z}}_{\text{DSC}}^i + \alpha \mathbf{F}_o^i(x_n), \quad (i = 1, 2) \tag{16}$$

where α is set to 0.1 empirically. Finally, the features from both branches are refined from the spectral perspective via $\mathcal{L}_{spa}$ and from the spatial perspective via $\mathcal{L}_{spe}$ to fine-tune spatial information and perform spectral correction

$$\mathcal{L}_{spa} = \|\nabla_{hw}\tilde{\mathbf{Z}}_{\text{DIV}}^1 - \nabla_{hw}\mathbf{Z}\|_1, \ \mathcal{L}_{spe} = \|\nabla_c \tilde{\mathbf{Z}}_{\text{DIV}}^2 - \nabla_c \mathbf{Z}\|_1 \tag{17}$$

where $\mathbf{Z}$ is ground truth (GT). ∇_{hw} and ∇_c stand for the gradient operator for spatial and spectral dimensions, respectively.

E. Training Objective

We apply a joint gradient loss constraint across the width, height, and channel dimensions. During training, the gradient backpropagation guides the model in generating images with enhanced edge and texture details while strengthening the spatial–spectral correlation between the potential HR-HSIs of the two branches. The $\mathcal{L}_{grad}$ is expressed as

$$\mathbf{G} = \|\mathbf{G}_c^2 + \mathbf{G}_h^2 + \mathbf{G}_w^2 + \varepsilon\|_2 \tag{18}$$

$$\mathcal{L}_{grad} = \frac{1}{n}\sum_{i=1}^{n} \|\mathbf{G}_{\hat{\mathbf{Z}}} - \mathbf{G}_{\hat{\mathbf{Z}}'}\|_1 \tag{19}$$

where $\mathbf{G}_c = \|\hat{\mathbf{Z}}_c - \hat{\mathbf{Z}}_{c-1}\|_1$, $\mathbf{G}_h = \|\hat{\mathbf{Z}}_h - \hat{\mathbf{Z}}_{h-1}\|_1$, and $\mathbf{G}_w = \|\hat{\mathbf{Z}}_w - \hat{\mathbf{Z}}_{w-1}\|_1$ are the gradients in the channel, height, and width dimensions, respectively. $G_{\hat{\mathbf{Z}}}$ and $G_{\hat{\mathbf{Z}}'}$ are the gradient norms of the reconstructed HR-HSI $\hat{\mathbf{Z}}$ and $\hat{\mathbf{Z}}'$ from the two branches. ε is a small constant for stability.

We choose the $\mathcal{L}1$ standard reconstruction loss

$$\mathcal{L}_{\hat{z}} = \frac{1}{n}\sum_{i=1}^{n} \|\hat{\mathbf{Z}} - \mathbf{Z}\|_1 \tag{20}$$

$$\mathcal{L}_{\hat{z}'} = \frac{1}{n}\sum_{i=1}^{n} \|\hat{\mathbf{Z}}' - \mathbf{Z}\|_1. \tag{21}$$

In summary, the total loss can be represented as

$$\mathcal{L}_{total} = \mathcal{L}_{\hat{z}} + \mathcal{L}_{\hat{z}'} + \lambda_1 \mathcal{L}_{grad} + \lambda_2 \mathcal{L}_{spa} + \lambda_3 \mathcal{L}_{spe} \tag{22}$$

where λ_1, λ_2, and λ_3 are hyperparameters of the equilibrium loss, empirically set to 0.1.

IV. Experiment

A. Experimental Settings

1) Datasets and Implementation Details: Our experiments utilize three datasets: CAVE, Harvard, and Chikusei. *CAVE dataset* [49] comprises 32 indoor images in size of 512 × 512 × 31. We select the first 20 HSIs for training, with the remaining 12 images designated for testing. *Harvard dataset* [50] comprises 50 indoor and outdoor images captured in daylight, each with a size of 1040 × 1392 × 31. We randomly choose 30 images as the training set, with the remaining 20 used as the test set. *Chikusei dataset* [51], captured by the Headwall Hyperspec-VNIR-C imaging sensor, consists of images with dimensions of 2517 × 2335 × 128. Due to the absence of edge information, we crop the central region of each image to 2048 × 2048 × 128 pixels, and then divide it into 16 nonoverlapping 512 × 512 × 31 images for experimentation. Randomly, 10 of these images are selected as the training set, with the remaining six images used as the test set. The LR-HSIs are generated through Gaussian blurring and by applying an 8× downsampling factor. Following the Wald protocol, HR-MSIs are produced using the Landsat-8 spectral response function for Chikusei and the Nikon D700 for CAVE and Harvard. Original HR-HSIs serve as GT, and all methods are evaluated under identical settings for fairness.

We implement our method under the PyTorch framework with NVIDIA GeForce RTX 4090 GPUs. We apply the Adam optimizer with a cosine annealing learning rate initialized to 2×10^{-4} in the training stage. The experiments are conducted in 250k iterations on all datasets. Unless otherwise specified, we use the same hyperparameter configuration across datasets.

2) Compared Methods and Evaluation Metrics: For comprehensive comparison, we select three traditional methods: GSA [34], FUSE [45], HySure [32] and seven deep learning-based methods: MHFNet [19], PSRT [23], HSR-Diff [46], MIMO-SST [26], M2DTN [38], Pan-Mamba [47], and FusionMamba [48]. We adopt four quality indices for quantitative evaluation [52]: peak signal-to-noise ratio (PSNR), structural similarity index (SSIM), spectral angle mapper (SAM), and relative global synthesis error (ERGAS).

B. Quantitative Comparisons

As shown in Table I, the proposed MosyMamba outperforms all SOTA algorithms across all evaluation metrics on the publicly available CAVE, Harvard, and Chikusei datasets. Note that the values in the table represent the average metrics of all testing samples. FusionMamba, MIMO-SST, and M2DTN represent the latest deep learning-based algorithms for HS/MS fusion. In terms of pixel-level metrics, MosyMamba achieves a PSNR that is 1.9027 dB higher and an SSIM that is 0.0013 higher than FusionMamba on the CAVE dataset, while its SAM and ERGAS are 0.526 and 0.1124 lower, respectively. The table clearly demonstrates that MosyMamba exhibits the most significant performance improvement on the CAVE dataset. This indicates that our method produces reconstructed HSIs with stronger spatial structures and fewer spectral distortions when handling scenes containing various real-world materials and objects. Since these four metrics respectively

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Fig. 4. Visual SR results and the corresponding error images (mean squared error) of different HS/MS fusion methods on the CAVE, Harvard, and Chikusei datasets. We display the images with bands 70–100–36 as R-G-B to show the Chikusei dataset.

TABLE I

AVERAGE QUANTITATIVE PERFORMANCE OF VARIOUS METHODS ON CAVE, HARVARD, AND CHIKUSEI DATASETS. AMONG THEM, GSA [34], FUSE [45], AND HYSURE [32] ARE TRADITIONAL ALGORITHMS. MHFNET [19], PSRT [23], HSR-DIFF [46], MIMO-SST [26], M2DTN [38], PAN-MAMBA [47], AND FUSIONMAMBA [48] ARE DEEP LEARNING-BASED ALGORITHMS. THE BEST RESULT IS HIGHLIGHTED IN BOLD; THE SECOND-BEST IS UNDERLINED. THE ↑ OR ↓ INDICATES HIGHER OR LOWER VALUES CORRESPONDING TO BETTER RESULTS

Methods	CAVE				Harvard				Chikusei			
	PSNR ↑	SSIM ↑	SAM ↓	ERGAS ↓	PSNR ↑	SSIM ↑	SAM ↓	ERGAS ↓	PSNR ↑	SSIM ↑	SAM ↓	ERGAS ↓
GSA [34]	40.8604	0.9654	3.7173	2.5457	37.7733	0.9533	6.4834	3.8519	32.9044	0.9121	6.9636	5.7303
FUSE [45]	31.2599	0.9487	8.2799	5.2845	28.8751	0.9224	12.0692	6.1995	26.3829	0.8773	8.4207	6.6885
HySure [32]	40.6326	0.9602	3.6971	2.3581	40.9232	0.9751	4.8448	2.8862	36.7171	0.9355	6.4014	5.0393
MHFNet [19]	44.1568	0.9873	2.8975	1.5584	41.6721	0.9863	3.2326	2.5902	39.5215	0.9689	1.8802	3.0468
PSRT [23]	45.8743	0.9905	2.8821	1.4955	42.9238	0.9876	2.9226	2.3609	40.0162	0.9732	1.7801	2.8016
HSR-Diff [46]	46.3618	0.9941	2.5089	1.1402	43.0876	0.9882	2.9181	2.2088	39.9508	0.9709	1.7903	2.8369
MIMO-SST [26]	46.7046	0.9949	2.4506	1.0036	43.4835	0.9899	2.8864	2.1469	40.3437	0.9756	1.6479	2.6292
M2DTN [38]	46.6568	0.9946	2.6625	1.1462	43.5066	0.9892	2.8421	2.1456	40.4148	0.9760	1.6495	2.5105
Pan-Mamba [47]	46.3945	0.9944	2.5607	1.1532	42.9506	0.9879	2.9203	2.3077	40.2926	0.9752	1.6537	2.6219
FusionMamba [48]	46.7658	0.9953	2.4722	0.9965	43.5781	0.9895	2.8469	2.1420	40.5878	0.9763	1.6406	2.5158
MosyMamba	**48.6685**	**0.9966**	**1.9462**	**0.8841**	**44.7988**	**0.9911**	**2.7459**	**1.9864**	**42.5068**	**0.9782**	**1.4956**	**2.1958**

characterize the spatial and spectral similarity between the predicted results and the GT, MosyMamba's predictions align most closely with the GT. This superiority stems from the proposed CIMB and DPFM modules, which fully interact with and preserve high-resolution spatial and spectral information. Such outstanding performance validates the effectiveness of MosyMamba.

Fig. 1 illustrates the relationship between PSNR and the number of parameters as well as flops. Compared with recent deep baseline models, including FusionMamba, MosyMamba achieves the best reconstruction accuracy while maintaining a compact model size. This advantage stems from the well-designed strategies adopted in the method. The proposed collaborative scanning mechanism divides a long token sequence of length L into multiple groups (9), reducing the maximum sequential length of each scanning group from L to M, thereby alleviating scanning latency on parallel hardware. As a result, MosyMamba

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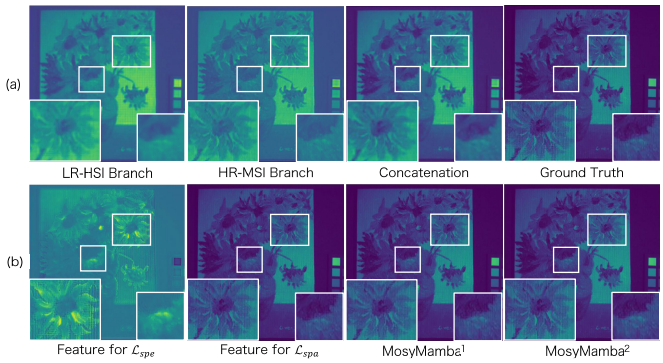


Fig. 5. Limit of existing methods and advantages of our method. The region within the white box is chosen for the analysis.

TABLE II
EFFECT OF MODULE COMPONENTS. THE ↑ OR ↓ INDICATES HIGHER OR LOWER VALUES CORRESPONDING TO BETTER RESULTS

CIMB	DPFM	PSNR ↑	SSIM ↑	SAM ↓	ERGAS ↓
✗	✗	37.5685	0.9403	2.6581	4.4569
✗	✓	40.6874	0.9738	1.6548	2.5962
✓	✗	41.2962	0.9766	1.5964	2.3505
✓	✓	**42.5068**	**0.9782**	**1.4956**	**2.1958**

provides a favorable cost–performance tradeoff for HS/MS fusion.

C. Qualitative Comparisons

As illustrated in Fig. 4, we present visual comparisons of different methods across three datasets. These visualization results demonstrate that traditional numerical methods yield inferior outcomes due to their reliance on handcrafted priors, which fail to fully capture HSI information. On the contrary, while deep learning-based methods achieve satisfactory results, the proposed MosyMamba excels in both detail preservation and global context. In particular, in the red-boxed areas highlighting key details—such as the dog's eyes and ears in the CAVE dataset, the streetlight in the Harvard dataset, and the realistic houses in the Chikusei dataset—MosyMamba produces sharper textures in both the RGB and residual images. Consequently, MosyMamba achieves outstanding performance both quantitatively and visually.

D. Ablation Study

1) Motivation Analysis: Fig. 5(a) highlights the main limitations of current feature fusion methods by comparing LR-HSI and HR-MSI outputs. The spectral intensity encoding patterns of LR-HSI features (visible through chromatic distributions) are not fully complementary to the spatial details of HR-MSI, leading to potential spectral occlusion in simple fusion approaches (e.g., concatenation), particularly in high-frequency regions such as petal edges. Fig. 5(b) displays feature maps under the FDF strategy and the network's outputs, demonstrating that our method effectively captures finer details in both spatial and spectral domains.

2) Component Analysis: Ablation studies on the Chikusei dataset assess each component's impact. As shown in Table II,

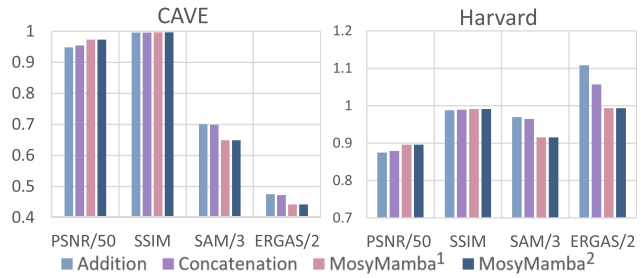


Fig. 6. Effect of DPFM in the proposed method. MosyMamba[1] and MosyMamba[2] denote the outputs of our method, respectively.

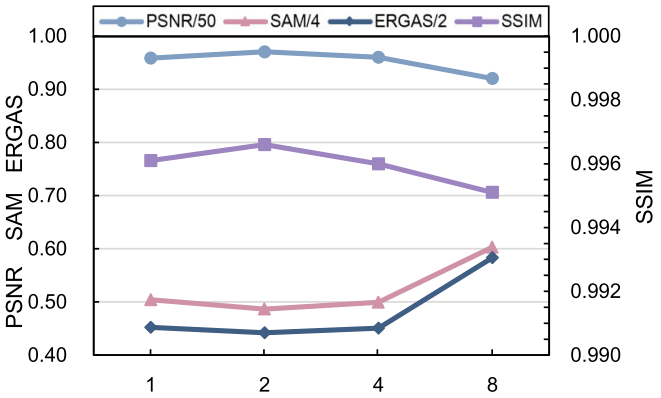


Fig. 7. Effect of various sampling strides in JSS2D.

DPFM alone demonstrates competitive performance even without CIMB for deep feature extraction. CIMB provides significant individual contributions, highlighting the effectiveness of incorporating surrounding spectral information and the synergistic scanning mechanism. Furthermore, the synergy between DPFM and CIMB further enhances the model's capability to recover fine details.

3) Effect of CIMB: The CIMB is designed to facilitate the interaction and extraction of information between LR-HSI and HR-MSI modalities. To evaluate the specific contribution of the proposed CIMB within the same overall framework, we keep all other components unchanged. In particular, we substitute CIMB with SW-MSA [53], DCTransformer [39], and Vmamba [28]. In addition, we remove CA and SA as another replacement strategy, since it does not support cross-modal interaction. Experimental results, as shown in Table III, clearly demonstrate that the proposed CIMB integrates more accurate information, leading to superior performance across the three datasets. Notably, replacing CIMB with a standard VMamba block causes a clear performance drop, indicating that the gain is not merely from using an SSM backbone, but from the proposed cross-modality coupling design.

4) Effect of DPFM: To validate the effectiveness of the proposed DPFM, we compare it against standard feature fusion methods, including direct addition and concatenation-convolution. Fig. 6 demonstrates the superior capability of DPFM in integrating complex dual-branch features, leading to significant performance improvements. Furthermore, the reconstructed image quality across MosyMamba's two outputs is virtually identical.

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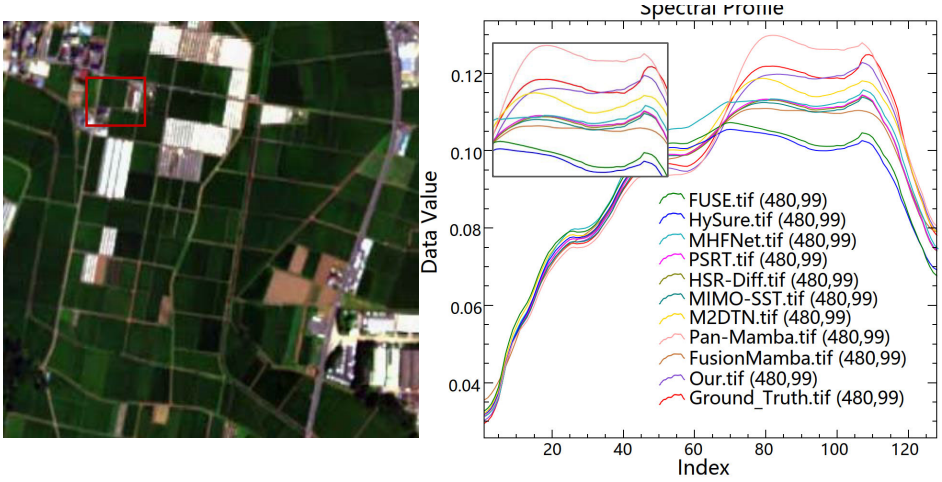


Fig. 8. Test image in Chikusei and the spectral profile of the area in the red rectangle are provided, respectively. The red line represents the GT, while the closest purple line corresponds to MosyMamba's results, demonstrating the superior capability of MosyMamba in recovering fine texture details.

TABLE III

EFFECT OF CIMB IN THE PROPOSED METHOD. THE ↑ OR ↓ INDICATES HIGHER OR LOWER VALUES CORRESPONDING TO BETTER RESULTS

Dataset	Method	PSNR ↑	SSIM ↑	SAM ↓	ERGAS ↓
CAVE	SW-MSA	46.2825	0.9939	2.4506	1.1012
	DCTransformer	46.6508	0.9945	2.5892	1.1358
	Vmamba	46.0429	0.9935	2.4637	1.1478
	w/o CA&SA	46.3621	0.9942	2.3766	1.1178
	MosyMamba	**48.6685**	**0.9966**	**1.9462**	**0.8841**
Harvard	SW-MSA	42.9296	0.9815	2.9013	2.4154
	DCTransformer	43.4915	0.9891	2.8726	2.1461
	Vmamba	42.9556	0.9853	2.9245	2.4568
	w/o CA&SA	43.1502	0.9806	2.9347	2.3925
	MosyMamba	**44.7988**	**0.9911**	**2.7459**	**1.9864**
Chikusei	SW-MSA	40.7706	0.9747	1.6395	2.5656
	DCTransformer	40.9239	0.9753	1.6054	2.4329
	Vmamba	40.8394	0.9752	1.6128	2.4950
	w/o CA&SA	41.2885	0.9767	1.5852	2.3415
	MosyMamba	**42.5068**	**0.9782**	**1.4956**	**2.1958**

5) Sampling Stride in JSS2D: The sampling stride in JSS2D affects the model's ability to capture regional features. We conduct experiments on the CAVE dataset to find the optimal stride. The results, normalized as shown in Fig. 7, excessively large sampling strides hinder the capture of close-range features. The best performance is achieved at stride = 2, which is aligned with the proposed odd-even sampling rule, validating the design choice of the synergistic scanning mechanism.

6) Visual Spectral Feature: As illustrated in Fig. 8, a randomly selected region (displayed by a red box) from the Chikusei dataset is used to compare the spectral feature performance of different methods. The right panel of Fig. 8 shows that our method (purple line) achieves the closest alignment to the GT, particularly in the higher wavelength bands, outperforming other baselines. This demonstrates our method's ability to effectively leverage the inherent high-resolution spectral characteristics of LR-HSI, enabling richer detail recovery.

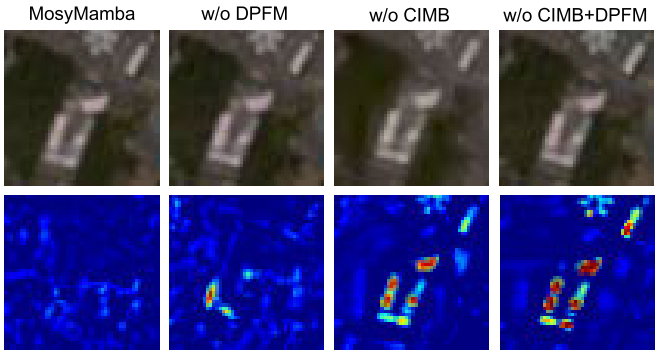


Fig. 9. Effect of module components on visual results.

TABLE IV

EFFECT OF LOSS FUNCTION COMPONENTS

$\mathcal{L}_{grad}$	$\mathcal{L}_{spa}$	$\mathcal{L}_{spe}$	PSNR ↑	SSIM ↑	SAM ↓	ERGAS ↓
✗	✗	✗	41.9683	0.9770	1.5339	2.2706
✓	✗	✗	42.1357	0.9772	1.5286	2.2510
✓	✓	✗	42.3565	0.9776	1.5175	2.2365
✓	✓	✓	**42.5068**	**0.9782**	**1.4956**	**2.1958**

7) Visual Results of Module Components: To investigate the impact of each model component, we conducted a componentwise ablation study, with Fig. 9 presenting visual results on the Chikusei dataset. The residual images visually demonstrate that the CIMB significantly enhances overall model performance by effectively restoring the image's spatial distribution and textures. Building upon this foundation, the DPFM further refines fine-grained textures and performs spectral correction, thereby improving the fidelity of the reconstructed results. These visual observations validate the effectiveness of each component in MosyMamba.

8) Effect of Loss Function: To investigate the effectiveness of additional components in the loss function, we conduct experimental analysis on the Chikusei dataset. As shown in Table IV, $\mathcal{L}_{grad}$, $\mathcal{L}_{spa}$, and $\mathcal{L}_{spe}$ significantly improve key

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TABLE V
SENSITIVITY ANALYSIS OF LOSS WEIGHTS ON THE CHIKUSEI DATASET. THE ↑ OR ↓ INDICATES HIGHER OR LOWER VALUES CORRESPONDING TO BETTER RESULTS

$\lambda_1 = \lambda_2 = \lambda_3$	PSNR ↑	SSIM ↑	SAM ↓	ERGAS ↓
0.001	42.3912	0.9778	1.5127	2.2314
0.1	**42.5068**	**0.9782**	**1.4956**	**2.1958**
0.2	42.4385	0.9779	1.5072	2.2176

metrics. $\mathcal{L}_{grad}$ encourages the two branch outputs to share consistent edge/texture strength across spatial and spectral gradients, while $\mathcal{L}_{spa}$ and $\mathcal{L}_{spe}$ explicitly refine the spatial and spectral perspectives. These results demonstrate that the proposed loss terms effectively enhance the refinement of spatial structures and spectral characteristics, thereby ensuring high-fidelity reconstruction performance.

9) Sensitivity Analysis of Loss Weights: To further investigate the influence of the loss weights, we conduct a sensitivity analysis by setting $\lambda_1 = \lambda_2 = \lambda_3 = \lambda$ and varying λ among 0.001, 0.1, and 0.2. As shown in Table V, the model achieves the best performance when $\lambda = 0.1$. When λ is too small, the constraints from L_{grad}, L_{spa}, and L_{spe} become insufficient, which limits the refinement of spatial and spectral details. When λ is too large, the model may overemphasize the auxiliary constraints and slightly affect the reconstruction fidelity. Therefore, we set $\lambda_1 = \lambda_2 = \lambda_3 = 0.1$ in all experiments.

10) Computational Complexity Analysis: Fig. 1 shows the comparison of PSNR, parameters, and flops. Compared with recent deep learning-based methods, including FusionMamba, MosyMamba achieves higher reconstruction accuracy with a compact model size and relatively low computational cost. This efficiency mainly benefits from the proposed synergistic scanning mechanism, which divides the long token sequence of length L into shorter groups of length M as formulated in (9), thereby reducing scanning redundancy and alleviating scanning latency.

V. CONCLUSION

In this article, we propose a novel modal-synergy Mamba network (MosyMamba) for hyperspectral and multispectral image fusion. The proposed MosyMamba utilizes a branch interaction approach to explore the unique and complementary information from LR-HSI and HR-MSI, resulting in high-fidelity HR-HSI. The customized CIMB significantly enhances the extraction and retention of LR-HSI spectral information and HR-MSI spatial details. The DPFM focuses on supplementing the spatial or spectral information for fine-grained feature refinement in each branch, with an FDF strategy to dynamically guide the detail recovery in regions with significant differences. Experiments on three public datasets demonstrate that our method achieves superior HSI reconstruction performance compared to SOTA methods.

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